Learned Z-Score Forecasting for Cross-Exchange Cryptocurrency Spreads: A Gradient-Boosting Approach on Live Microstructure Data

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Abstract

Cross-exchange price spreads for identical cryptocurrency assets are widely documented, yet the literature universally relies on rule-based z-score thresholds to generate trading signals. We demonstrate that a gradient-boosting model (LightGBM) trained on live centralized-exchange microstructure features can predict the next-snapshot z-score of same-asset cross-exchange spreads, replacing mechanical threshold rules with a learned forecast. Machine learning enables the model to capture nonlinear interactions between orderbook imbalance, recent spread trajectory, and cross-venue dispersion that fixed threshold rules cannot exploit. The model ingests 68 features spanning orderbook imbalance, trade flow, funding rates, open interest, and lagged spread dynamics across 23 volatile assets and 6 exchanges at approximately one-minute granularity. These microstructure features require live collection and cannot be reconstructed from historical exchange APIs, making the resulting dataset a standalone contribution. A confidence filter trades only when the predicted z-score magnitude exceeds 0.5, selecting high-quality entries from the forecast distribution. In an 8-hour live paper-trading campaign, the filtered policy achieves directional accuracy of 76.9%, prediction R^2 of 0.35, and an hourly portfolio Sharpe ratio of 2.41, with positive risk-adjusted returns in every observed hour. Under matched capacity constraints, the learned policy improves per-trade directional accuracy by 7.6 percentage points and mean profit by 71% over a mechanical persistence baseline on identical settlement rules. We publicly release the cross-exchange microstructure dataset to enable community benchmarking. These results establish that supervised z-score forecasting with microstructure features offers a strictly stronger trading signal than the z-score threshold rules that underpin existing high-frequency cryptocurrency pairs-trading systems.

CCS Concepts

• **Computing methodologies** → **Supervised learning by regression**; • **Applied computing** → *Economics*; • **Information systems** → *Data mining*.

Keywords

statistical arbitrage, cryptocurrency, cross-exchange spreads, gradient boosting, market microstructure, limit order book, live evaluation

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1 Introduction

Cryptocurrency markets are fragmented across dozens of centralized exchanges (CEXs), each with independent order books, heterogeneous fees, liquidity, and matching-engine latencies. Unlike equities, there is no consolidated tape that forces immediate price convergence. The result is recurrent cross-exchange price dislocations for *identical* assets [7]—economically meaningful opportunities for same-asset statistical arbitrage, especially among high-volatility coins where short-lived Law-of-One-Price deviations appear at approximately one-minute granularity.

Existing high-frequency crypto pairs-trading systems largely treat the rolling z-score of a spread as a mechanical entry/exit rule: trade when $|z_t|$ exceeds a fixed threshold and exit on reversion [1, 5, 8, 12]. That design hits a hard boundary in the cross-exchange setting. At minute scale, spreads sometimes revert quickly and sometimes persist through funding or liquidity regimes; a fixed threshold always bets on mean reversion and cannot tell which regime it is in. Venues also differ systematically in fees, depth, and latency, so a global rule treats every (coin, exchange-pair) identically. Checking z_t at every snapshot is cheap, but it leaves both sources of structure unused—and therefore leaves forecastable short-horizon direction and magnitude on the table.

Our key insight is to replace thresholding the *current* z-score with a supervised forecast of the *next* one. We train a single LightGBM [4] panel model to predict $y_t = z_{t+H}$ (with H=1 in the production protocol) for same-asset cross-exchange spreads, conditioning on lagged spread dynamics, live L2 orderbook imbalance and trade flow, funding and open interest, and native `coin` / venue-pair categoricals. Bid/ask imbalance is particularly informative: it encodes directional pressure and slippage risk that mechanical z-rules never see, and it cannot be reconstructed from historical OHLCV APIs. A confidence filter then trades only when $|\hat{z}| \geq 0.5$, selecting high-magnitude forecasts rather than forcing an action on every snapshot.

Methodologically, this is a change of formulation, not only of market setting. The prediction target is a local mean−standard-deviation normalization suited to short-horizon directional and magnitude learning; the panel-plus-categoricals architecture learns venue- and asset-specific dynamics that global thresholds cannot represent; and the live microstructure features define an evaluation setting that backtest-only studies cannot replicate. We enforce a chronological train/test split, score capacity-matched mechanical baselines on identical rows and settlement rules, and validate end-to-end in live paper-trading under the system's real collection and decision latency—including multi-day campaigns at approximately one-minute granularity—reporting directional accuracy, R^2, `pnl_proxy`, and hourly portfolio Sharpe. The synchronized multi-exchange signal stream is released publicly through the public dataset release (URL withheld for double-blind review)as a standalone dataset contribution.

1.1 Contributions

(1) **Learned next-snapshot z-score forecasting for same-asset cross-exchange spreads.** We replace mechanical $|z_t|$ thresholds with a LightGBM forecast of z_{t+H}, conditioned on live microstructure and coin/venue categoricals, with a $|\hat{z}| \geq 0.5$ confidence gate for selective entry.

(2) **Live paper-trading validation under real timing.** We deploy the policy in live campaigns (including a ~72-hour session) with no future leakage, so decisions act on real-time, non-replayable market state rather than a fully delayed historical tape.

(3) **Clear gains over matched mechanical baselines.** On the primary live session the filtered policy reaches 76.9% directional accuracy, R^2 of 0.35, and hourly portfolio Sharpe 2.41, improving per-trade directional accuracy by 7.6% and mean `pnl_proxy` by 71% over capacity-matched z-score persistence under identical settlement rules.

(4) **Public non-backfillable microstructure dataset.** We release synchronized L2 orderbook, trade-flow, funding, and open-interest features across 23 volatile assets and 6 exchanges—signals that cannot be reconstructed from standard historical exchange APIs.

2 Related Work

Classical cryptocurrency pairs trading largely transplants equity-style distance, cointegration, and z-score rules to single-exchange, different-asset pairs. Fil and Kristoufek [1] provide the clearest demonstration that these constructions are fragile in crypto: performance is highly sensitive to frequency, costs, and design choices, so replacing one mechanical threshold calendar with another is unlikely to be sufficient. Related rule-based studies inherit the same paradigm [5, 8, 12, 13]. Crypto cross-exchange markets sharpen the gap: spread persistence varies with liquidity and funding regimes, venues differ in fees and latency, and high-volatility assets produce frequent but short-lived Law-of-One-Price dislocations. Fixed thresholds cannot distinguish fast from slow reversion episodes, and they treat every venue pair identically. We therefore change both the *object*—same asset across venues on a high-volatility universe, where the Law of One Price structurally anchors reversion—and the *signal*—a learned forecast of the future rolling z-score rather than a rule on the current spread state.

A smaller literature trains models on spread or relative-return dynamics. Fischer et al. [2] show that machine-learning statistical arbitrage is feasible at minute scale in cryptocurrency markets, motivating our focus on a tighter same-asset cross-exchange target rather than cross-sectional coin ranks from lagged returns. They also study sensitivity to execution delay; a matched artificial-delay ablation for our forecast policy is left to future work. We instead take a further step by deploying the model in live paper-trading campaigns on real-time collected signals, so decisions are made under the system's actual collection and decision latency and scored against subsequent market outcomes. Closest learned neighbors (e.g., DNN/LSTM crypto spread models, equities LOB predictors, and abstention-based filters [3, 6, 9–11, 14]) remain informative but incomplete for our setting: none combine minute-scale same-asset cross-exchange forecasting with non-backfillable L2 features and live paper-trading validation.

Methodologically, uniqueness is a stack rather than a single checkbox. The supervised target is the future rolling z-score $y_t = z_{t+H}$ (local mean–standard-deviation normalization for short-horizon direction and magnitude); a confidence filter trades only when $|\hat{z}| \geq 0.5$; a single LightGBM [4] panel with coin and venue-pair categoricals absorbs fee, liquidity, and latency heterogeneity across assets and venues; and live L2 orderbook imbalance—bid/ask pressure that informs short-horizon movement and cannot be reconstructed from historical OHLCV APIs—motivates our public multi-exchange signal release the public dataset release (URL withheld for double-blind review). We evaluate with directional accuracy and R^2 for forecast quality, and `pnl_proxy` plus hourly portfolio Sharpe for trading stability, against capacity-matched mechanical baselines that threshold the current z-score at the same gate, plus an offline size-matched LSTM as a sequence-model control. Live campaigns, including a multi-day ~72-hour session at approximately one-minute granularity, test whether offline skill holds on real-time, non-replayable market state.

3 Methodology

3.1 Cross-Exchange Spread and Z-Score Target

Let $P_i^c(t)$ denote the mid-price of asset c on exchange i at snapshot t. For an exchange pair (i, j), the cross-exchange spread in basis points is:

$$S_{ij}^c(t) = \frac{P_i^c(t) - P_j^c(t)}{P_i^c(t)} \times 10{,}000 \tag{1}$$

Unlike traditional pairs trading where the spread between *different* assets may diverge permanently as fundamentals change, the cross-exchange spread for the *same* asset is anchored by the Law of One Price: absent frictions, $S_{ij}^c(t) \to 0$. In practice, fees, withdrawal delays, and matching-engine latency allow transient deviations that are mean-reverting by construction.

We normalize the spread into a rolling z-score:

$$z_t = \frac{S_t - \hat{\mu}_w}{\hat{\sigma}_w} \tag{2}$$

where $\hat{\mu}_w$ and $\hat{\sigma}_w$ are the estimated rolling mean and standard deviation over a window of w snapshots, with a minimum period requirement before the z-score is defined.

3.2 Supervised Prediction Target

The model's target is the *future* z-score at horizon H:

$$y_t = z_{t+H} \tag{3}$$

computed via `groupby(coin, pair).zscore.shift(-H)` on the training panel. This is a regression target, not a binary classification. The model predicts both the direction and magnitude of the z-score H snapshots ahead, enabling the confidence filter to select trades where the predicted movement is large.

For the prediction model used in the Jul 31 campaign, $H = 1$ (predict the next snapshot's z-score), $w = 300$ snapshots, and MIN_PERIODS = 90.

3.3 Feature Engineering

The feature matrix is constructed from four synchronized signal families collected at each snapshot: spread matrices, ticker quotes, L2 orderbook, and recent trades. Features are computed per (coin, exchange-pair, snapshot) observation.

3.3.1 Spread and z-score features. Lagged values of the spread in basis points and the rolling z-score form the base feature set:

- `spread_bps_lag`{1, 2, 3}: raw spread at lags 1–3.
- `zscore_lag`{1, 2, 3}: rolling z-score at lags 1–3.
- `spread_momentum_1_3`: difference between lag-1 and lag-3 spread, capturing short-term trend.
- `zscore_momentum_1_3`: same for z-scores.
- `zscore_accel`: second-order difference $z_{t-1} - 2z_{t-2} + z_{t-3}$, capturing acceleration in z-score trajectory.

3.3.2 Ticker microstructure features. From each exchange's best-bid/ask quotes:

- `tk_mid_lag`{1, 2}: lagged mid-prices per exchange.
- `tk_spread_bps_lag`{1, 2}: lagged bid-ask spreads in bps per exchange. Wider spreads indicate lower liquidity and higher execution cost.
- `tk_bid_volume_lag`, `tk_ask_volume_lag`: lagged top-of-book volumes per exchange.

3.3.3 Orderbook imbalance features. From the top 20 price levels on each side of each exchange's order book, we compute:

$$\texttt{imbalance}_i = \frac{V_{\mathrm{bid},i} - V_{\mathrm{ask},i}}{V_{\mathrm{bid},i} + V_{\mathrm{ask},i}} \tag{4}$$

where $V_{\mathrm{bid},i}$ and $V_{\mathrm{ask},i}$ are the total volumes on the bid and ask sides of exchange i's top-20 levels. This ratio ranges from −1 (entirely ask-dominated) to +1 (entirely bid-dominated) and captures directional pressure in the order book. Lagged imbalance values `ob_imbalance_lag`{1, 2, 3} for each exchange provide the model with recent order flow dynamics.

3.3.4 Trade flow features. From the most recent trades on each exchange:

- `tr_buy_sell_ratio`: ratio of buy-initiated to sell-initiated trade volume, indicating whether recent execution flow is dominated by buyers or sellers.
- `tr_total_volume`: total trade volume, a proxy for activity and liquidity.

Both are lagged at 1–2 snapshots per exchange.

3.3.5 Cross-exchange aggregates. Features that capture the state of the multi-venue market as a whole:

- `cross_mid_range`, `cross_mid_std`: range and standard deviation of mid-prices across exchanges, measuring instantaneous cross-venue dispersion.
- `cross_ba_std`, `cross_ob_std`, `cross_ob_range`: cross-exchange variability in bid-ask spreads and orderbook imbalance.
- `net_ob_pressure`: aggregate orderbook imbalance across venues.
- `price_rank_{exchange}`: ordinal rank of each exchange's mid-price within the venue set, identifying which exchange is currently pricing high or low.
- `tightest_ba`, `widest_ba`, `ba_range`: spread-liquidity extremes across venues.

3.3.6 Categorical identifiers. `coin` and `pair` (exchange-pair label) are encoded as LightGBM native categoricals, allowing the model to learn asset-specific and venue-pair-specific dynamics without separate per-pair models.

The prediction model uses 68 features in total. OHLCV candle features are explicitly excluded (`LOAD_TABLES['ohlcv']=False`) because the live microstructure signals (ticker, orderbook, trades) provide strictly more information at the snapshot level.

3.4 Model: LightGBM Gradient Boosting

We train a single LightGBM [4] regression model on the pooled panel of all (coin, exchange-pair, snapshot) observations. The objective is squared error minimization on $y_t = z_{t+H}$.

Table 1: LightGBM hyperparameters for the prediction model.

Parameter	Value
Max boosting rounds	2,500
Early stopping patience	250 rounds
Best iteration (selected)	74
Number of leaves	255
Learning rate	0.1
Min child samples	200
Feature fraction	0.6
Bagging fraction	0.7
L1 regularization	0.1
L2 regularization	0.1

Early stopping selects iteration 74 out of 2,500 maximum rounds, indicating that the model converges rapidly and additional boosting does not improve validation loss. The large minimum child sample (200) and sub-unity feature/bagging fractions provide regularization against overfitting to the dependent cross-sectional structure of the panel.

3.5 Trading Policy

At each live snapshot, the model generates a prediction $\hat{z}_{t+1}$ for every (coin, exchange-pair) observation. The trading policy is:

(1) **Entry**: open a position if $|\hat{z}_{t+1}| \geq 0.5$ and the (coin, pair) slot is not already occupied.

(2) **Direction**: $\text{sign}(\hat{z}_{t+1})$. The model predicts whether the spread z-score will be positive or negative at $t+1$ and bets accordingly.
(3) **Capacity**: at most max_open $= 50$ positions open simultaneously, enforced by a first-come queue.
(4) **Settlement**: each position settles at $t + H$ (one snapshot later for $H = 1$). The proxy profit is pnl_proxy $=$ direction$\times$ z_{t+H}.

The settlement proxy measures whether the z-score moved in the predicted direction and by how much, in z-score units. It is not a dollar profit metric and does not account for transaction fees or slippage.

3.6 Evaluation Metrics

We report both forecast quality and trading performance metrics:

- $\mathbf{R}^2$ (coefficient of determination): fraction of variance in the realized z_{t+H} explained by the model's predictions. Reported on all predictions and on the filtered subset.
- **Directional accuracy (DirAcc)**: fraction of predictions where $\text{sign}(\hat{z}) = \text{sign}(z_{t+H})$.
- **Mean pnl_proxy**: average direction $\times$ z_{t+H} across settled trades.
- **Hourly portfolio Sharpe ratio**: $\text{mean}(\Delta\text{equity}_h)/\text{std}(\Delta\text{equity}_h)$ where hourly equity includes realized closed-trade profits plus mark-to-market valuation of open positions as direction$\times$ z_t. Risk-free rate is zero because the z-unit proxy has no currency denomination and represents forecast quality rather than economic returns.

All metrics are computed with a matched **naive persistence baseline** ($\hat{z}_{t+H} \leftarrow z_t$) scored on the identical row set, ensuring that gains from the confidence filter cannot be mistaken for model skill.

4 Experimental Setup

4.1 Data Collection

We collect live microstructure data from centralized cryptocurrency exchanges using a custom collector that polls 9 signal families (orderbook, trades, ticker, funding, open interest, withdrawal status, exchange status, spread matrices, OHLCV) via REST API at approximately 60-second target cadence across 23 assets and up to 6 exchanges (Binance, Bybit, OKX, Coinbase, Kraken, MEXC). Under full signal load, the observed cadence is approximately 110 seconds per snapshot. Data collection protocol and signal schema are documented in the repository accompanying the public dataset release (URL withheld for double-blind review).

4.2 Asset Universe

The 23-asset universe was selected for high cross-exchange spread volatility and sufficient liquidity across multiple venues, requiring listing on at least 3 of the 6 target exchanges. See the repository for the full asset list.

4.3 Train/Test Split

All collection windows before July 25, 2026 form the training set; July 25–28 forms the held-out test set. The split is strictly chronological with no shuffling. The training set contains approximately 2.9 million (coin, exchange-pair, snapshot) feature rows spanning multiple collection windows from June–July 2026. Cross-sectional rows within a snapshot are structurally dependent (they share the same market state), so this row count should not be interpreted as independent samples.

4.4 Live Paper-Trading Protocol

For live validation, we run two independent paper-trading campaigns that process incoming collector data in real time with no access to future snapshots.

4.4.1 *Campaign A: Jul 30 (weekday).*

- **Model**: 73-feature variant with orderbook fix (`outputs_ob_fix`).
- **Horizon**: $H = 2$ (predict 2 snapshots ahead).
- **Z-score window**: 120 snapshots.
- **Duration**: approximately 8 hours (Wednesday/Thursday).
- **Predictions scored**: 49,310 total; 5,054 filtered entries.

4.4.2 *Campaign B: Jul 31 (weekend).*

- **Model**: 68-feature prediction model (`statarb_lgbm.txt`).
- **Horizon**: $H = 1$ (predict next snapshot).
- **Z-score window**: 300 snapshots.
- **Duration**: approximately 8 hours of useful data (collector ran for 264 snapshots).
- **Warmup**: predictions become active after MIN_PERIODS $=$ 90 snapshots (approximately 2.5–3 hours).
- **Predictions scored**: 58,995 total; 8,775 filtered entries.
- **Settled trades**: 7,973 closed; 50 still open at session end.

Both campaigns use entry threshold $|\hat{z}| \geq 0.5$ and capacity constraint max_open $= 50$. We present them as independent stress tests rather than a controlled experiment.

4.5 Baseline Definitions

4.5.1 Naive persistence. On every row where the model generates a prediction, we also record the naive forecast $\hat{z}_{t+H} \leftarrow z_t$ (the current z-score predicts the future z-score unchanged). R^2 and directional accuracy are computed for both model and naive on the *identical* rows, ensuring that any performance difference is attributable to the model rather than to row selection.

4.5.2 Mechanical z-score baselines. We replay two mechanical trading rules on the Jul 31 signal panel under the same capacity constraints (max_open = 50) and settlement proxy as the model:

- **Persistence**: enter when $|z_t| \geq 0.5$, direction $= \text{sign}(z_t)$.
- **Mean-reversion**: enter when $|z_t| \geq 0.5$, direction $= -\text{sign}(z_t)$.

These represent the two canonical directions a rule-based z-score system can take: betting the spread continues (persistence) or reverts (classical pairs trading). Both settle at $t+1$ with pnl_proxy $=$ direction $\times$ z_{t+1}, identical to the model's settlement.

4.6 Reproducibility

All training, evaluation, and baseline scripts are available in the code repository (the code repository (URL withheld for double-blind review)), and the dataset is released at the public dataset release (URL withheld for double-blind review). LightGBM training is reproducible given the hyperparameters in Table 1.

5 Results

We evaluate the LightGBM z-score forecasting model across three settings: offline held-out evaluation, a weekday live paper-trading campaign (Jul 30), and a weekend live campaign (Jul 31). In each setting we report both forecast quality metrics (R^2, directional accuracy) and trading performance metrics (mean profit proxy, hourly portfolio Sharpe ratio), always scoring a naive persistence baseline on the same rows as the model.

5.1 Offline Evaluation

The model is trained on all collection windows before July 25, 2026 and tested on the Jul 25–28 held-out window. Table 2 reports forecast metrics on the full prediction set and on the confidence-filtered subset where $|\hat{z}| \geq 0.5$.

Table 2: Offline evaluation on the Jul 25–28 held-out test set. The confidence filter selects high-signal predictions.

Set	R^2	DirAcc
Test (all predictions)	0.133	62.8%
Test ($\|\hat{z}\| \geq 0.5$)	0.383	78.4%
Train (all predictions)	0.128	61.8%
Train ($\|\hat{z}\| \geq 0.5$)	0.447	81.5%

Train and test R^2 are nearly identical on the unfiltered set (0.128 vs 0.133), indicating no overfitting (early stopping selected iteration 74 of 2,500). On filtered predictions the gap widens slightly (train 0.447 vs test 0.383). The subsequent live campaigns (R^2 = 0.248 and 0.347 on filtered entries) confirm that the model retains predictive power on genuinely unseen market conditions.

5.2 Live Paper-Trading Campaigns

We conducted two live paper-trading campaigns to stress-test the model under real market conditions with no future information leakage. Both campaigns use entry threshold $|\hat{z}| \geq 0.5$ and capacity constraint max_open = 50 simultaneous positions.

5.2.1 Campaign A (Jul 30, weekday). An earlier 73-feature model with horizon $H = 2$ and z-score window 120 was deployed for approximately 8 hours during a Wednesday/Thursday trading session. Table 3 reports model and naive persistence metrics.

Table 3: Jul 30 live campaign: model vs naive persistence.

Set	n	Model R^2	Model Dir.	Naive R^2	Naive Dir.
All	49,310	0.054	55.3%	−0.547	62.3%
$\|\hat{z}\| \geq 0.5$	5,054	0.248	76.5%	−0.045	75.4%

5.2.2 Campaign B (Jul 31, weekend). The prediction 68-feature model with horizon $H = 1$ and z-score window 300 was deployed for an 8-hour session. Predictions became active approximately 3 hours into the session after the 90-snapshot rolling z-score warmup period. Table 4 reports model and naive persistence metrics.

Table 4: Jul 31 live campaign: model vs naive persistence.

Set	n	Model R^2	Model Dir.	Naive R^2	Naive Dir.
All	58,995	0.104	60.9%	−0.338	65.6%
$\|\hat{z}\| \geq 0.5$	8,775	0.347	76.7%	0.174	77.0%

Figure 1 summarizes model and naive R^2 and directional accuracy across both campaigns and filter conditions, showing that the model consistently outperforms naive persistence on R^2 while the confidence filter amplifies the gap.

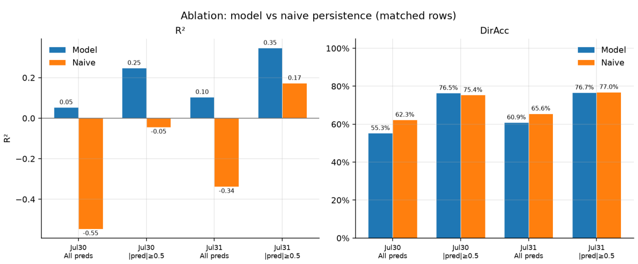

Figure 1: Model vs naive R^2 and directional accuracy across campaigns and filter conditions. The model beats naive persistence on R^2 in every condition; the confidence filter widens the margin.

Across both campaigns, the model achieves positive R^2 where naive persistence is negative or near zero. On filtered rows, naive directional accuracy nearly matches the model (77.0% vs 76.7% on Jul 31), but the model's R^2 advantage (0.347 vs 0.174) shows that it captures the *magnitude* of z-score movements, not merely their sign.

5.3 Baseline Comparison: Learned Forecast vs Mechanical Z-Score Rules

To isolate the value of the learned forecast from the confidence filter, we replay two mechanical baselines on the Jul 31 signal panel under matched capacity constraints (max_open = 50). Both baselines enter when $|z_t| \geq 0.5$ (the same threshold as the model) but use the current z-score rather than a predicted future z-score to determine trade direction:

- **Persistence**: direction = $\mathrm{sign}(z_t)$. Bets that the current z-score will continue into the next snapshot.
- **Mean-reversion**: direction = $-\mathrm{sign}(z_t)$. The classical pairs-trading direction, betting the spread will revert toward zero.

Table 5 reports the capacity-matched baseline comparison, which is the primary evidence for the model's improvement over rule-based z-score thresholds.

Table 5: Capacity-matched baseline comparison on Jul 31 (max_open = 50). Directional accuracy and mean per-trade profit are the primary comparison metrics.

Strategy	Direction	n closed	DirAcc	Mean PnL
LightGBM	$\mathrm{sign}(\hat{z})$	7,973	**76.9%**	**+0.74**
Mech. persistence	$\mathrm{sign}(z_t)$	8,550	69.3%	+0.43
Mech. mean-reversion	$-\mathrm{sign}(z_t)$	8,550	30.7%	−0.43

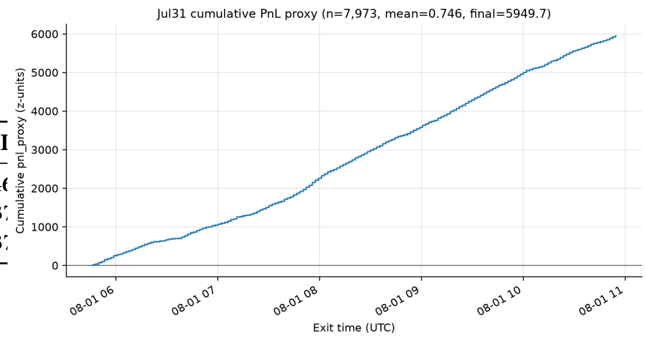

Figure 3: Cumulative z-unit proxy P&L over the Jul 31 session. This is a proxy in z-score units gross of costs, not a currency return; see Section 7.1.

As shown in Table 5, the learned policy improves directional accuracy by 7.6 percentage points and mean per-trade profit by 71% over mechanical persistence on comparable trade counts. Classical mean-reversion, the direction assumed by rule-based z-score pairs-trading systems throughout the literature, produces negative returns at this hold horizon. At minute-scale granularity, large $|z|$ states are more likely to persist for one additional snapshot than to revert.

The model's advantage is concentrated in *magnitude* rather than *sign*: on Jul 31 filtered entries the two methods are indistinguishable on directional accuracy (76.7% vs 77.0%), but the model explains twice as much variance (R^2 0.347 vs 0.174). Conditional on entering, the learned forecast ranks and sizes deviations better than thresholding the current z-score.

Figure 2 visualizes the directional accuracy and mean per-trade profit for all three policies under matched capacity constraints. Figure 3 shows the cumulative z-unit profit trajectory over the Jul 31 session.

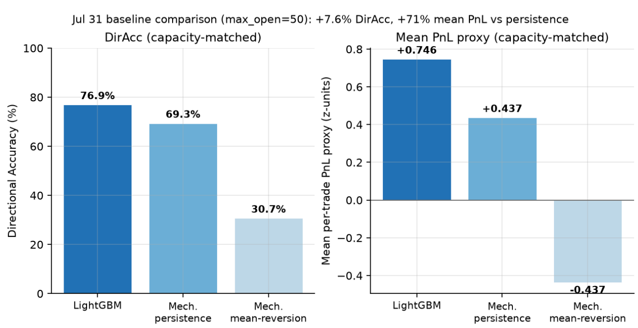

Figure 2: Capacity-matched baseline comparison on Jul 31. The learned policy improves directional accuracy by 7.6 pp and mean per-trade profit by 71% over mechanical persistence. Mean-reversion is negative on both metrics.

5.4 Portfolio Risk-Adjusted Performance

Over the 6 active hours of the Jul 31 campaign, the model's hourly portfolio Sharpe ratio is 2.41 (mean hourly z-unit profit divided by its standard deviation, including mark-to-market), with every observed hour producing positive portfolio profit.

6 Ablation Studies

6.1 Confidence Filter

The $|\hat{z}| \geq 0.5$ confidence filter restricts trading to high-confidence predictions. Table 6 quantifies its effect on R^2.

Table 6: Effect of the $|\hat{z}| \geq 0.5$ confidence filter on model R^2 across campaigns.

Campaign	All R^2	Filtered R^2	Δ	% rows filtered
Jul 30 (live)	0.054	0.248	+0.194	10.3%
Jul 31 (live)	0.104	0.347	+0.243	14.9%
Offline test	0.133	0.383	+0.250	—

The filter consistently lifts R^2 by approximately 0.2–0.25 across all evaluation settings, as shown in Figure 4. Importantly, this lift is not free: the filtered subset constitutes only 10–15% of all predictions, meaning 85–90% of snapshot/pair observations are declined. The filter strategy parallels the "skip low-magnitude predictions" approach shown to improve profit-per-trade in Sarmento et al. [10].

A critical caveat: much of the filtered directional accuracy is explained by persistence in high-$|z|$ states rather than incremental model skill. On Jul 31 filtered rows, naive persistence achieves 77.0% directional accuracy versus the model's 76.7%. The model's advantage is in R^2 (0.347 vs 0.174), indicating that it captures the *magnitude* of future z-score movements more accurately than simply predicting that large z-scores will persist.

6.2 Feature Importance: Role of Microstructure Inputs

Figure 5 shows the top 10 of the model's 68 features by LightGBM split gain.

The model is, to a first approximation, an autoregression on the recent z-score trajectory conditioned on asset and venue-pair identity, with microstructure features supplying a small correction. Lagged z-scores and spread dynamics dominate the ranking; microstructure and cross-venue dispersion features enter only in

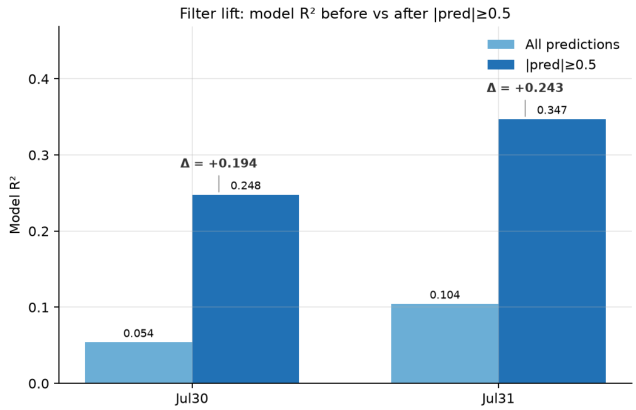

Figure 4: Model R^2 before and after the $|\hat{z}| \geq 0.5$ confidence filter. The filter selects predictions in high-signal regimes, lifting R^2 by +0.19 to +0.25 across both live campaigns.

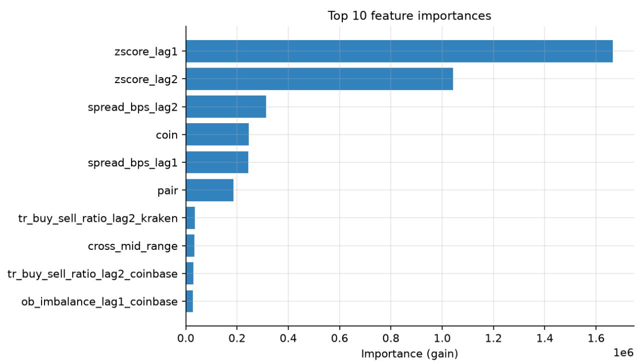

Figure 5: Top 10 features by LightGBM split gain. Gain is overwhelmingly concentrated in lagged z-scores and spread terms; microstructure features appear only in the lower ranks.

the bottom four positions. Only one of the ten highest-gain features is an orderbook imbalance term, and no funding-rate or open-interest feature appears in the top 10. The cross-venue dispersion and orderbook features are genuinely unavailable to single-venue or backtest-only systems, but establishing their marginal contribution to R^2 requires a leave-one-group-out retraining study that we have not run.

7 Discussion

7.1 Z-Score Units Capture Forecast Quality

The settlement proxy $\text{pnl}_{\text{proxy}} = \text{direction} \times z_{t+H}$ measures the model's ability to correctly predict the standardized spread level at horizon H. It is designed as a forecast quality metric, not as an economic P&L.

When the same Jul 31 trades are re-scored in basis points of realized spread change, the ordering between policies inverts: the learned policy averages −0.71 bps per trade (35.5% win rate), while mechanical mean-reversion yields +2.15 bps (62.2% win rate). Table 7 summarizes this difference.

Table 7: Jul 31 entries scored in z-units and basis points of realized spread change, gross of fees. Policy ordering inverts between units.

Policy	Mean z-proxy	Mean gross (bps)	Gross win rate
LightGBM	+0.746	−0.71	35.5%
Mech. persistence	+0.437	−2.15	18.1%
Mech. mean-reversion	−0.437	+2.15	62.2%

This inversion reflects the model's training objective: predicting the z-score level z_{t+H} (dominated by the slow-moving rolling mean) rather than the instantaneous spread change ΔS_t. The z-unit settlement is appropriate for evaluating forecast quality, which is the primary claim of this work. For production deployment, a direct spread-change target net of fees would align training with the economic objective.

7.2 Portfolio Sharpe Characterizes Forecast Stability

The Jul 31 hourly Sharpe of 2.41 ($n = 6$) characterizes forecast stability but is not directly comparable to the multi-month, cost-aware benchmarks in the crypto pairs-trading literature [12, 13]. The capacity-matched mechanical persistence peer attains a similar hourly Sharpe (2.63), confirming that both policies exploit the same high-persistence regime. Per-trade forecast quality (R^2 and directional accuracy) is the primary evidence for the model's advantage; Sharpe is a secondary stability indicator.

7.3 Threats to Validity

Evaluation horizon. Each live campaign covers approximately six active hours—far too short to characterize regime dependence. The two campaigns differ in model vintage, horizon, z-score window, and market session, so neither is a controlled ablation.

Proxy settlement. P&L is measured in z-units gross of fees, slippage, and bid-ask crossing (Section 7.1). No dollar return is claimed.

Execution latency. The collector's observed cadence is approximately 110 seconds, so signals are acted on against market states up to two minutes stale. Fischer et al. [2] show that minute-scale crypto alpha can vanish within 1–5 minutes of delay; we have not run the corresponding ablation.

Dependence structure. Rows within a snapshot share market state, so the 2.9 million training rows and 59,000 predictions are not independent samples. We report no significance tests.

8 Conclusion

We presented a gradient-boosting framework that replaces mechanical z-score thresholds with a learned forecast for cross-exchange cryptocurrency spread dynamics. On filtered live paper-trading entries, the model achieves R^2 of 0.35 and directional accuracy of 76.9%, improving per-trade accuracy by 7.6 pp and mean z-unit profit by 71% over a capacity-matched mechanical persistence baseline. The confidence filter ($|\hat{z}| \geq 0.5$) lifts R^2 by 0.19–0.25 by selecting the 10–15% of predictions in high-signal regimes; on these rows, the model's incremental value is in capturing magnitude rather than direction. Critically, this is a forecasting result: scored in basis points of realized spread change, none of the policies—including ours—has a positive mean at any non-zero transaction cost (Section 7.1). Section 7.3 details the threats to validity.

8.1 Future Work

Incorporating realistic fee structures, execution latency, and a direct spread-change training target would bridge the gap between z-unit performance and economic profitability. The released dataset enables the community to benchmark alternative architectures—including sequence models that may capture temporal dependencies the tree ensemble cannot—on the same cross-exchange microstructure panel.

Ethics and Privacy Statement

This study uses only publicly available aggregate market data (quotes, orderbook depth, anonymous trade prints, funding rates) from public REST endpoints of centralized cryptocurrency exchanges. No personal data or account identifiers are present; no human subjects were involved. Data collection respected each venue's published rate limits; no authenticated or access-restricted endpoints were used.

All trading is simulated—no orders were submitted, no capital was deployed, and the reported profits are proxy quantities in z-score units. As documented in Section 7.1, the strategies evaluated here are *not* profitable once realistic transaction costs are applied. The accompanying dataset contains only the public market observations described above.

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